解答例

$$
\begin{aligned}
&\int \frac{x^2}{(x\sin x+\cos x)^2}\,dx \\
=&\int \frac{x\cos x\cdot\frac{x}{\cos x}}{(x\sin x+\cos x)^2}\,dx \\
=&\int \frac{x\cos x}{(x\sin x+\cos x)^2}\cdot\frac{x}{\cos x}\,dx \\
=&\int \frac{(x\sin x+\cos x)'}{(x\sin x+\cos x)^2}\cdot\frac{x}{\cos x}\,dx \\
=&-\frac{1}{x\sin x+\cos x}\cdot\frac{x}{\cos x}-\int -\frac{1}{x\sin x+\cos x}\cdot\left(\frac{x}{\cos x}\right)'\,dx \\
=&-\frac{1}{x\sin x+\cos x}\cdot\frac{x}{\cos x}+\int \frac{1}{x\sin x+\cos x}\cdot\left(\frac{x}{\cos x}\right)'\,dx \\
=&-\frac{1}{x\sin x+\cos x}\cdot\frac{x}{\cos x}+\int \frac{1}{x\sin x+\cos x}\cdot\left(\frac{\cos x-(-x\sin x)}{\cos^2 x}\right)\,dx \\
=&-\frac{1}{x\sin x+\cos x}\cdot\frac{x}{\cos x}+\int \frac{1}{x\sin x+\cos x}\cdot\left(\frac{x\sin x+\cos x}{\cos^2 x}\right)\,dx \\
=&-\frac{1}{x\sin x+\cos x}\cdot\frac{x}{\cos x}+\int \frac{1}{\cos^2 x}\,dx \\
=&-\frac{1}{x\sin x+\cos x}\cdot\frac{x}{\cos x}+\tan x+C
\end{aligned}
$$

ただし C は積分定数